

4010

B.C.A. IInd Semester Examination, 2025

DATA STRUCTURE

Paper : NBCA-201

Time : 2 Hours]

[M.M. : 70

Note :- Attempt *Four* questions in all. *One* question out of two questions (10 marks each) is to be attempted from Section 'A'. Remaining *three* questions (each carrying 20 marks) are to be attempted out of eight questions given in Section 'B'.

Section-A

[10×1=10]

Note :- Attempt any *one* question.

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(1)

KP-550

Turn Over

1. Define the singly and doubly linked list. Also show how insertion at the beginning and deletion at the end are handled in doubly linked list.

Or

2. Write a C program to implement a circular queue using array (include operation for enqueue, dequeue and display). Discuss condition for queue overflow and underflow, and explain how the circular nature of the queue resolves the limitation of linear queue.

Section-B

[20×3=60]

Note :- Attempt any *three* questions.

3. Explain the following sorting algorithms in C with suitable example of each :

(i) Insertion sort

(ii) Selection sort

For each, analyze the best, average, and worst case time complexity. Also find best, average and worst case complexity for the array {29, 10, 2, 14, 37, 14} using selection sort algorithm.

4. Compare and contrast linear search and binary search techniques. Write a C program to implement both. Demonstrate how binary search reduce time complexity. Also, derive the number of comparison needed in the worst case.

5. Differentiate between strictly Binary tree, complete Binary tree and extended Binary tree with examples. Construct a Binary tree from the given Inorder and Pre-order traversal :

Inorder : A B C D E M N O P Q R S T

Pre-order : M B A D C E Q O N P S R T

Show the step by step construction and final tree structure.

6. Convert the following infix expression to postfix express.

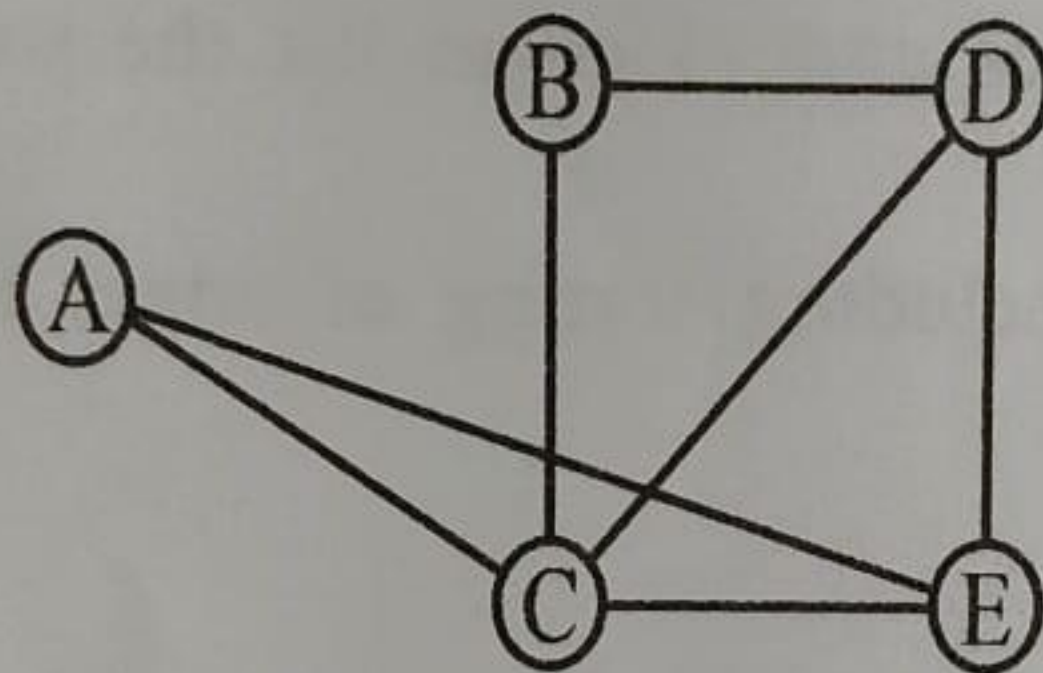
Show step by step conversion using stack :

Infix : $((6 * 2 / 3 - 4) - 7 ^ 2 ^ 4 / 6 + (3 * 2))$

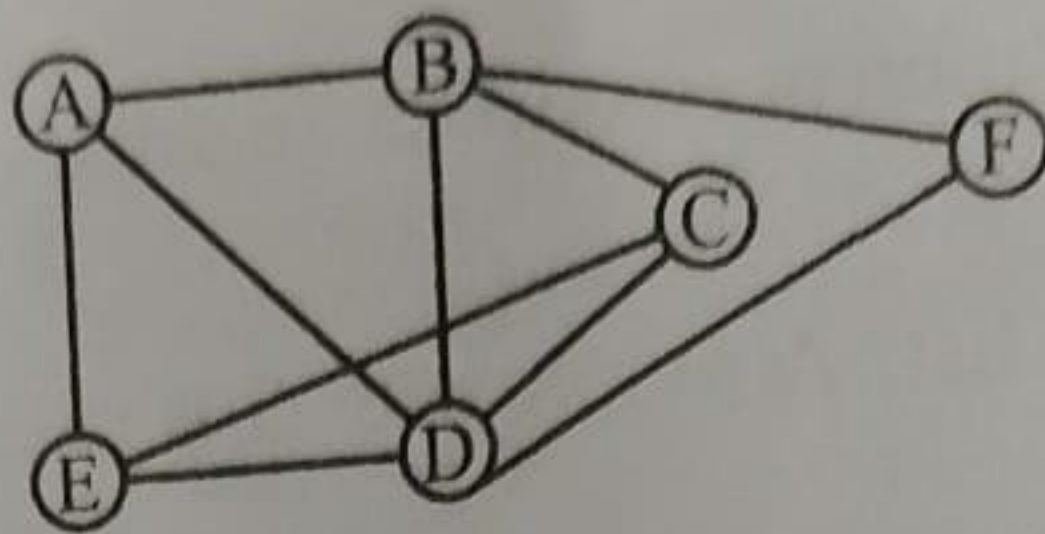
Also evaluate the resultant postfix expression using stack

to get the final result. (symbol represent power function)

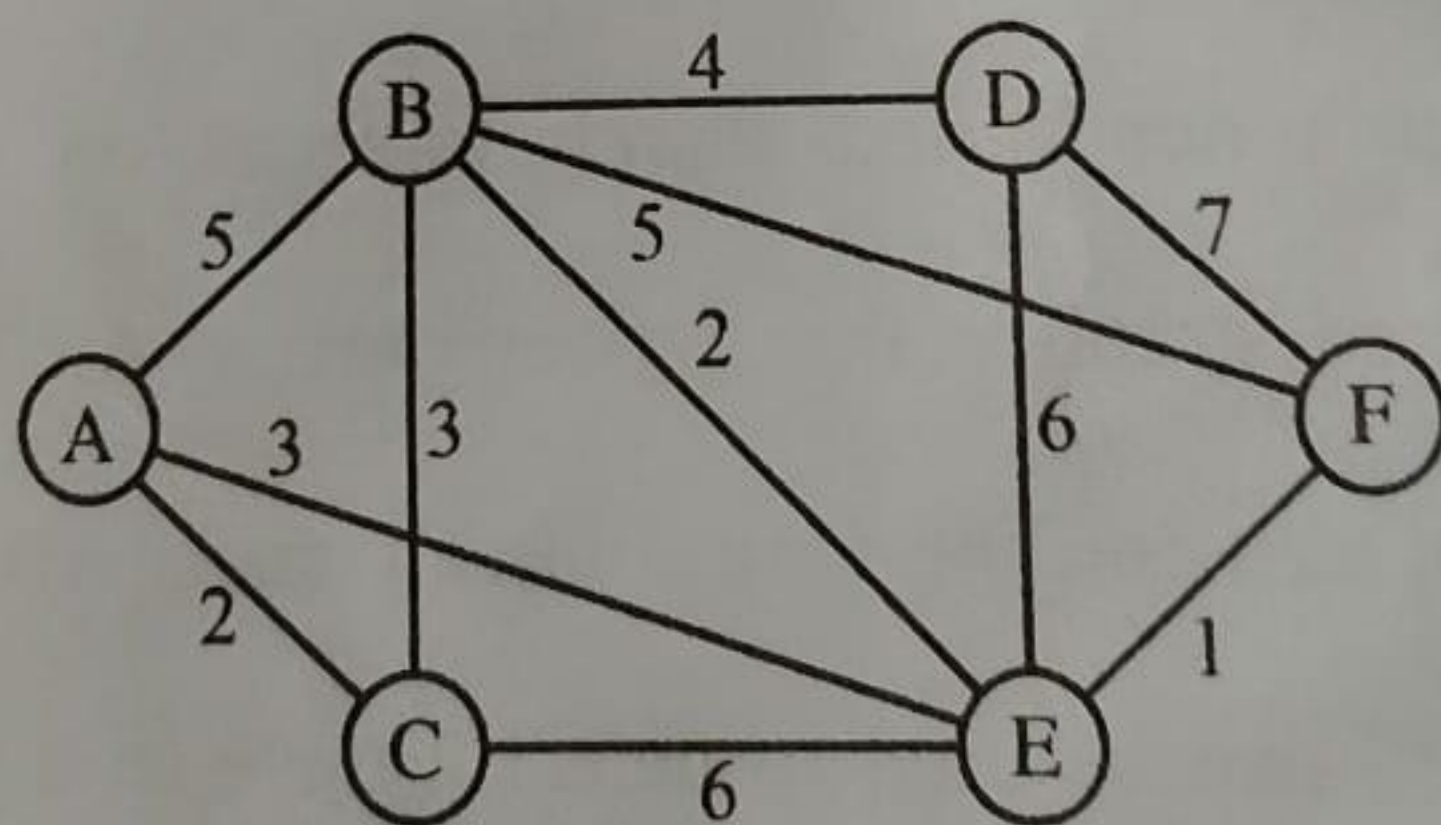
7. Explain the difference between an adjacency matrix and an adjacency list. A graph with 5 vertices and 6 edges, represent it using both method and compare their space efficiency. Which representation is better for sparse graphs and why ?



8. Perform a depth first search traversal on the given undirected graph starting from vertex A. Track the visited nodes and the recursion call stack. Also, explain how DFS can be used to detect a cycle in a graph :



9. Discuss spanning tree with example. Explain the working of Kruskal's algorithm for finding a minimum cost spanning tree. Given a weighted undirected graph with 6 vertices and 11 edges list the step of Kruskal's algorithm, including sorting of edges and union find operations :



10. Explain the derivation of index formulas for 1D, 2D and 3D arrays. Write C program for inserting and deleting elements from a 2D array. Discuss memory allocation and access patterns in multi-dimensional arrays.